Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 16: State Classification

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16.1 Transient and Recurrent States

Example 16.1 (Gambler's Ruin). *Suppose for simplicity that $N = 4$ in Example 1.1, so that the transition probability is*

	0	**1**	**2**	**3**	**4**
0	1.0	0	0	0	0
1	0.6	0	0.4	0	0
2	0	0.6	0	0.4	0
3	0	0	0.6	0	0.4
4	0	0	0	0	1.0

Question 1. *Compute p^n for a few values of n. What does that tell us about the qualitative nature of the states?*

Solution 1.

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}, \qquad p^{200} \approx \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 57/65 & 0 & 0 & 0 & 8/65 \\ 45/65 & 0 & 0 & 0 & 20/65 \\ 27/65 & 0 & 0 & 0 & 38/65 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Question 2. *Are we guarenteed to return to 0 and 4? Infinitely often?*

Solution 2. *Since $p(0,0) = 1$, it is an absorbing state, the chains come back on the next step with probability 1,*

$$\mathbb{P}_0(T_0 = 1) = 1.$$

This means $\rho_{00} = 1$, so we know that it is recurrent.

Since $p(4,4) = 1$, it is an absorbing state, the chains come back on the next step with probability 1,

$$\mathbb{P}_4(T_4 = 1) = 1.$$

This means $\rho_{44} = 1$, so we know that it is recurrent.

Question 3. *Can we start at 1 and return to 1? (Hint: think about $p^2(1,1)$).*

Solution 3.

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}$$

Yes, the easiest such example is I have to go from $1 \to 2 \to 1$*, which happens with* $p(1,2)p(2,1) = 0.4 \cdot 0.6 = 0.24$.

Question 4. *Are we guaranteed to start at 1 and return to 1?*

Solution 4. *If the chain goes to 0, it will never return to 1, so the probability of starting at 1 and not returning is*

$$\mathbb{P}_1(T_1 = \infty) \geq p(1,0) = 0.6 > 0.$$

Of course there are other ways to make this happen, but this is the easiest.

Question 5. *Are we guaranteed to start at 3 and return to 3?*

Solution 5. *If the chain goes to 4, it will never return to 3, so the probability of starting at 3 and not returning is*

$$\mathbb{P}_3(T_3 = \infty) \geq p(3,4) = 0.4 > 0.$$

Question 6. *Are we guaranteed to start at 2 and return to 2?*

Solution 6. *If the chain goes to 0, it will never return to 2, so the probability of starting at 2, going to 1, going to 0, and not returning is*

$$\mathbb{P}_2(T_2 = \infty) \geq p(2,1)p(1,0) = 0.6 \times 0.6 = 0.36 > 0.$$

Another way we can show that we can start at 2 and never return to 2 is below: If the chain goes to 4, it will never return to 2, so the probability of starting at 2, going to 3, going to 4, and not returning is

$$\mathbb{P}_2(T_2 = \infty) \geq p(2,3)p(3,4) = 0.4 \times 0.4 = 0.16 > 0.$$

Lastly, by looking at $p^2(i,j)$ *we see the very same answers:*

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}$$

Example 16.2. *We recall the transition probability of the social mobility chain:*

	Lower	**Middle**	**Upper**
Lower	0.7	0.2	0.1
Middle	0.3	0.5	0.2
Upper	0.2	0.4	0.4

To determine which states we return to we have to use a slightly different argument.

The reason is: unlike last time, we can get from any state to any other state. We don't have any absorbing states.

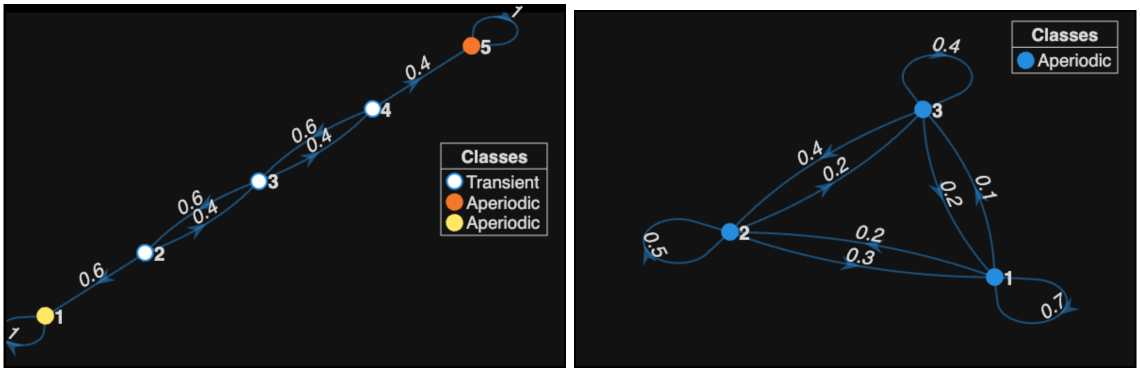

Figure 16.1: Comparing Chains to understand returning

To begin we note that no matter where X_n is, there is a probability of at least 0.1 of hitting 3 on the next step so

$$\mathbb{P}_3(T_3 > n) \leq 0.9^n$$

Since $0.9 \leq 1.0$, *then as we go further and further, we see that this* $(0.9)^n \to 0$ *as* $n \to \infty$.

This proves that the probability of starting at Upper class (state 3) and not returning in the first n generations becomes very rare. So this means you have to return, since those are two mutually exclusive events.

The same argument can be formulated for $\mathbb{P}_i(T_i > n) \leq 0.8^n$ *for* $i = \{1, 2\} = \{lower,\ middle\}$. *THis means all the states are recurrent. You have to return, and by the (strong) Markov property, you will return infinitely often.*

Definition 16.3. *Let*

$$T_y = \min\{n \geq 1 : X_n = y\}$$

be the time of the first return to y (i.e., being there at time 0 doesn't count) and let

$$\rho_{yy} = \mathbb{P}_y(T_y \leq \infty)$$

be the probability X_n returns to y when it starts at y.

Definition 16.4. *Let*

$$T_y^k = \min\{n > T_y^{k-1} : X_n = y\}$$

be the time of the k^{th} return to y. Remember starting at y and returning is ρ_{yy}, and by the strong Markov property, we know that the time for k returns is

$$\mathbb{P}_y(T_y^k \leq \infty) = \rho_{yy}^k.$$

This means one of two things:

- $\rho_{yy} < 1$: The probability of returning k times is $\rho_{yy}^k \to 0$ as $k \to \infty$ Thus, eventually the Markov chain does not find its way back to y. In this case the state y is called *transient*, since after some point it is never visited by the Markov chain.

- $\rho_{yy} = 1$: The probability of returning k times $\rho_{yy}^k = 1$, so the chain returns to y infinitely many times. In this case, the state y is called *recurrent*, it continually recurs in the Markov chain.

Definition 16.5. *We say x* communicates with y *and write $x \to y$ if there is a positive probability of reaching y from x:*

$$\rho_{xy} = \mathbb{P}_x(T_y < \infty) > 0.$$

Question 7. *Can we chain these properties together? What does this say about $x \to y$ and $y \to z$?*